

BioMTL

Multi-Task PPG Benchmark, v1.0

BioMTL is a PPG-centric benchmark for multi-task physiological monitoring. Each 4 s fingertip-PPG window sampled at 1000 Hz is associated with labels for seven tasks: systolic and diastolic blood pressure, blood glucose, heart rate, BMI, mental stress, hypertension, and type-2 diabetes.

Item	Value
Subjects	172
PPG windows	1,808
Signal	Fingertip PPG, 1000 Hz, 4 s windows
Tasks	4 continuous regression tasks and 3 binary classification tasks
Access	Restricted to non-commercial academic research under the Data Use Agreement

Archive contents

Path	Description
data/signals.npy	PPG windows, float32 array with shape (1808, 4000).
data/window_subject.npy	Subject id for each PPG window; row-aligned to signals.npy.
data/labels.csv	Subject-level labels and demographics, 172 rows.
data/example_window.csv	Rounded human-readable example of the first PPG window.
documents/en/DATASET_DOCUMENTATION.pdf	Detailed schema, processing, label dictionary, limitations, and citation information.
documents/en/DATA_USE_AGREEMENT.pdf	Terms of use and application form.

Access and citation

Access is limited to non-commercial academic research. Complete the Data Use Agreement and send the signed PDF to **biomtl@163.com**. Publications or public outputs using BioMTL should cite the associated publication:

Zhang, Z., Lu, H., & Zhao, Q. (2026). *PPGPT: Transferring Next-Token Modeling from Language to PPG Signals*. Proceedings of the AAAI Conference on Artificial Intelligence, 40(34), 28573-28581. <https://doi.org/10.1609/aaai.v40i34.40088>

Quick start

```
import numpy as np, pandas as pd
X = np.load("data/signals.npy")          # (1808, 4000) float32
sid = np.load("data/window_subject.npy")  # subject id per window
L = pd.read_csv("data/labels.csv").set_index("subject_id")
y_bg = L.loc[sid, "bg"].values           # per-window blood glucose label
```

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